

COVER SHEET, DRAWING INDEX & LEGEND

Electric batch oven - 18 kW heater bank, 2 x recirculation blowers, PLC / HMI control retrofit

1. DESIGN BASIS

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| Incoming supply  | 415 V AC, 3-phase + N + PE, 50 Hz, TN-S                                                        |
| Prospective Isc  | 10 kA rms sym. at panel incomer                                                                |
| Control supply   | 230 V AC ex 500 VA control transformer -T1                                                     |
| DC supply        | 24 V DC, 5 A regulated SMPS -G1                                                                |
| Heater load      | 18 kW total - 3 x 6 kW elements, star, 25 A/ph                                                 |
| Blower load      | 2 x 1.5 kW, 415 V, 3-ph, 3.5 A FLC, DOL start                                                  |
| Heater switching | Contactor -K3, driven from PLC output Y2                                                       |
| Temp. control    | Panel PID controller -A2, relay output OUT1                                                    |
|                  | wired as a volt-free signal to PLC input X6                                                    |
| Over-temperature | Independent high-limit thermostat -B3 wired directly in the -K3 coil circuit (hard-wired trip) |
| Door interlock   | Position switch -B1 to PLC input X5 only (software stop of blowers - see NOTE 4)               |
| PLC              | MELSEC iQ-F FX5U-32MT/ES (16 DI / 16 DO, sink)                                                 |
| HMI              | GOT2000 GT2107-WTBD 7" TFT, 24 V DC, Ethernet                                                  |
| PLC <-> HMI link | Ethernet 100BASE-TX, MELSOFT connection                                                        |
| Security         | Two separate passcodes: MAINTENANCE and QUALITY                                                |
| Cure timers      | Held while the oven is not fit to cure                                                         |
| Door watchdog    | Chatter or no 'door closed' signal for 10 min<br>-> alarm + volt-free output Y12 / -KA4        |
| Data logging     | FX5U SD card, 7-day rolling history, CSV                                                       |
| Enclosure        | Wall-mounted 800 x 600 x 250 mm, IP54, RAL 7035                                                |
| Standards        | IEC 60204-1, IEC 61439-1/-2, IS 8623                                                           |

2. CONTROL FUNCTIONS IMPLEMENTED IN THE PLC

- a) Blower 1 / Blower 2 start-stop from panel push-buttons and from the HMI.
- b) Door interlock: blowers stop while the door is open, heater stays ON; blowers restart automatically when the door is closed again.
- c) Heater contactor -K3 follows the PID controller demand signal (X6), gated by PLC permissives (E-stop, overloads, high-limit, no fault).
- d) Heat-up watchdog alarm: the demand contact X6 stays closed continuously for longer than the preset time, i.e. the oven never reaches set point and the controller never drops out => 'FAILURE TO REACH SETPOINT'.
- e) Door-signal watchdog: a chattering door switch, or no 'door closed' signal for 10 minutes, raises an alarm and energises the volt-free remote alarm output Y12 / -KA4 on terminals -X1:50 / -X1:51.
- f) Six independent retentive 2-hour slot timers (slot 1...6). Once started a timer runs to 02:00:00 and can only then be reset. An earlier reset needs the MAINTENANCE passcode. A cure timer is held, and does not run on to completion, whenever the oven is not fit to cure.
- g) Door open / door close event counters, separate and resettable.
- h) Lots loaded / lots unloaded counters derived from slot-timer start and reset events, with a mismatch indication.
- i) Two separate passcodes, not two levels: MAINTENANCE (early slot reset, manual test, settings) and QUALITY (resetting the counters). Neither role inherits the other's rights - see NOTE 11.
- j) Alarm and event history logged to the FX5U SD card, 7 days rolling, exportable to a USB stick from the HMI.

6. REVISION HISTORY

| REV | DATE       | DESCRIPTION                                                                         | BY     |
|-----|------------|-------------------------------------------------------------------------------------|--------|
| 0   | 2026-09-06 | First issue - schematic for review                                                  | CLAUDE |
| 1   | 2026-09-06 | Door-signal watchdog output Y12 / -KA4; 7-day event logging to the SD card          | CLAUDE |
| 2   | 2026-09-06 | -S0 contact to PLC X0 corrected to 21-22 (11-12 is used in the coil bus, sheet 05)  | CLAUDE |
| 3   | 2026-09-06 | Separate maintenance / quality passcodes; cure timers held when the oven is not fit | CLAUDE |

3. DRAWING INDEX

| SH. | DRAWING No.    | TITLE                                         |
|-----|----------------|-----------------------------------------------|
| 01  | OVN-2026-01-01 | Cover sheet, drawing index & legend           |
| 02  | OVN-2026-01-02 | Single line diagram - power distribution      |
| 03  | OVN-2026-01-03 | Power circuit - blower motors -M1 / -M2       |
| 04  | OVN-2026-01-04 | Power circuit - heater bank -E1/-E2/-E3       |
| 05  | OVN-2026-01-05 | Control supply, E-stop & 24 V DC distribution |
| 06  | OVN-2026-01-06 | PLC -A1 : CPU supply & digital inputs X0-X17  |
| 07  | OVN-2026-01-07 | PLC -A1 : digital outputs Y0-Y17              |
| 08  | OVN-2026-01-08 | Contactor coil circuits & interlock provision |
| 09  | OVN-2026-01-09 | Temperature controller & over-temp protection |
| 10  | OVN-2026-01-10 | HMI -A3, Ethernet network & door layout       |
| 11  | OVN-2026-01-11 | Panel general arrangement & gland plate       |
| 12  | OVN-2026-01-12 | Terminal strip schedule -X1 / -X2             |
| 13  | OVN-2026-01-13 | I/O schedule & bill of materials              |

4. LEGEND (IEC 60617)

|  |                                |  |                                   |
|--|--------------------------------|--|-----------------------------------|
|  | Contact, normally open         |  | Circuit breaker, thermal-magnetic |
|  | Contact, normally closed       |  | Switch-disconnector / isolator    |
|  | Push-button, NO, spring return |  | Contactor, main pole              |
|  | Push-button, NC, spring return |  | Thermal overload relay            |
|  | Emergency stop, latching, NC   |  | Motor, 3-phase asynchronous       |
|  | Position (door) switch, roller |  | Resistive heating element         |
|  | Contactor / relay coil         |  | Control transformer               |
|  | Indicator lamp, 24 V DC        |  | AC/DC power supply unit           |
|  | Audible alarm (hooter)         |  | Thermocouple, type K              |
|  | Fuse                           |  | Terminal / protective earth       |

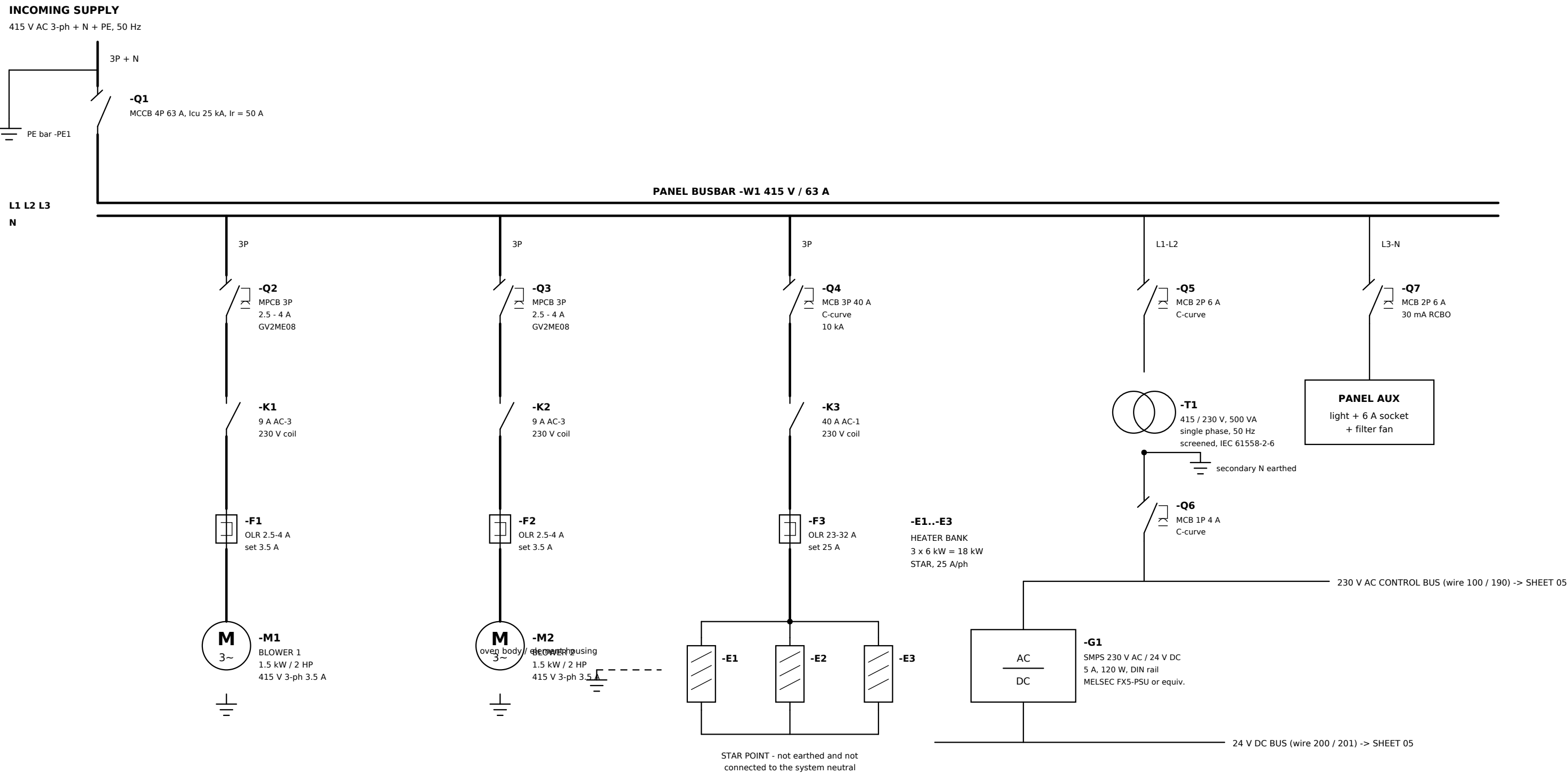
5. NOTES

1. All wiring to IEC 60204-1. Power conductors 1100 V grade PVC copper, control conductors 1.5 mm2 black, 24 V DC control conductors 1.0 mm2 dark blue, protective conductor green/yellow, neutral light blue.
2. All field wiring terminates on terminal strip -X1 / -X2. No field cable is landed directly on a device.
3. Heater over-temperature protection is HARD-WIRED: high-limit thermostat -B3 (manual reset) is in series with the -K3 coil, so the heater drops out on over-temperature regardless of the state of the PLC.
4. DOOR INTERLOCK - the door position switch -B1 is wired to PLC input X5 only, as instructed. The blowers are therefore stopped by the PLC program (software), not by hardware. A wire link -LK1 is fitted in the blower coil circuit on sheet 08; if a hard-wired interlock is required later, remove -LK1 and land the NC contact of a second (safety) door switch on terminals -X1:60 / -X1:61. No panel rework is needed.
5. The door interlock deliberately does NOT break the heater circuit: the heater stays energised while the door is open, per the process requirement. Warning label to be fitted at the door: 'HOT SURFACES - HEATER LIVE'.
6. PLC digital inputs are wired SINK type: terminal S/S is connected to +24 V and the field contact switches 0 V into the X terminal. PLC digital outputs are transistor SINK type: the load sits between +24 V and the Y terminal, COMO / COM1 are connected to 0 V. Contactor coils (230 V AC) are driven via interposing relays.
7. All fail-safe field contacts (E-stop, high-limit, stop push-buttons) are wired normally CLOSED so that a broken wire produces the safe state. Overload trip contacts are wired NO (input ON = tripped) for diagnostics.
8. 0 V of the 24 V DC system is bonded to PE at one point only, at terminal -X2:0V/PE in the panel.
9. Cable screens of the thermocouple extension lead are earthed at the panel end only.
10. Ratings shown are for the assumed loads listed in section 1. Confirm motor nameplate FLC and heater element rating on site and adjust -Q2/-Q3 settings, -F1/-F2/-F3 overload settings and cable sizes accordingly.
11. SEPARATION OF DUTIES - maintenance performs the early slot-timer resets; the quality team, and only the quality team, can clear the counters that record them, so the record cannot be erased by the person who made it. Both passcodes are held in the PLC, not the HMI, and must be changed from their defaults at commissioning.

|                                                    |        |       |            |                        |         |     |
|----------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER  |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>COVER SHEET, DRAWING INDEX & LEGEND |        |       |            |                        |         |     |
| DRAWN                                              | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                            | -      | SCALE | NTS        | OVN-2026-01            | 01 / 13 | 3   |

SINGLE LINE DIAGRAM - POWER DISTRIBUTION

415 V AC, 3-ph + N + PE, 50 Hz, TN-S, 10 kA



NOTES:

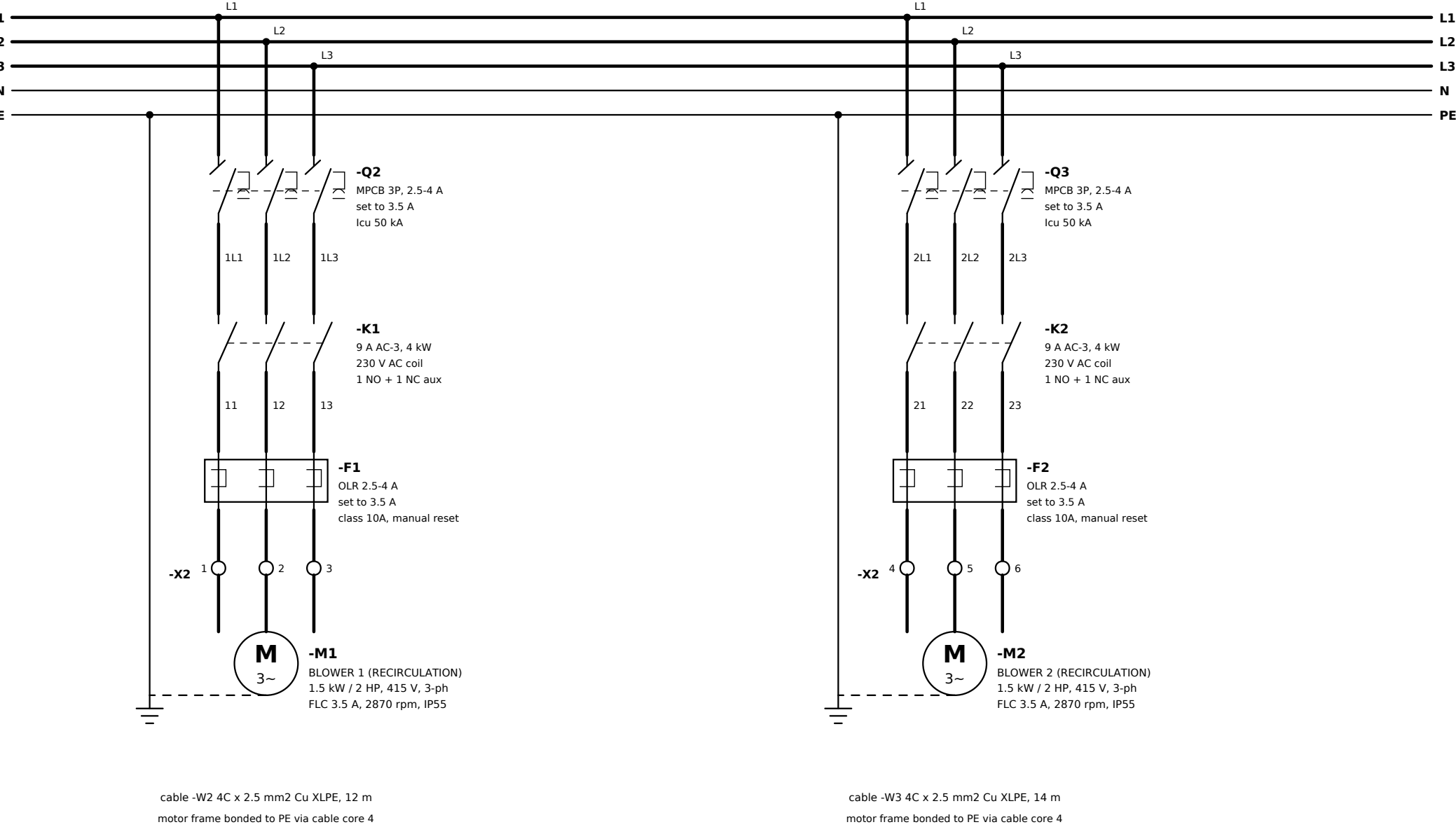
- 1. Diagram is single-line; all switching devices are 3-pole unless noted. See sheets 03 and 04 for the 3-line detail.
- 2. Heater elements are connected in STAR; the star point is not connected to the system neutral.
- 3. Cable schedule: -M1/-M2 4C x 2.5 mm2 Cu; heater bank 4C x 10 mm2 Cu; incomer 4C x 16 mm2 Cu.
- 4. -Q4 is a C-curve MCB sized for the resistive load; -F3 gives the overload/asymmetry protection for the elements.
- 5. -K3 is rated for AC-1 utilisation. Expect contact wear proportional to the PID cycle time - see NOTE 6 on sheet 04.

|                                                         |        |       |            |                        |         |     |
|---------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER       |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>SINGLE LINE DIAGRAM - POWER DISTRIBUTION |        |       |            |                        |         |     |
| DRAWN                                                   | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                 | -      | SCALE | NTS        | OVN-2026-01            | 02 / 13 | 3   |

POWER CIRCUIT - BLOWER MOTORS -M1 / -M2

Direct-on-line starters, 415 V 3-ph, 1.5 kW each

FROM PANEL BUSBAR -W1 (SHEET 02) 415 V AC, 3-ph + N + PE, 50 Hz



AUXILIARY CONTACT ALLOCATION (for reference only)

| DEVICE | CONTACT    | USED FOR                              |
|--------|------------|---------------------------------------|
| -K1    | 13-14 (NO) | PLC input X14 - run feedback sheet 06 |
| -K1    | 21-22 (NC) | spare<br>-                            |
| -K2    | 13-14 (NO) | PLC input X15 - run feedback sheet 06 |
| -K2    | 21-22 (NC) | spare<br>-                            |
| -F1    | 95-96 (NC) | in series with -K1 coil sheet 08      |
| -F1    | 97-98 (NO) | PLC input X10 - OL tripped sheet 06   |
| -F2    | 95-96 (NC) | in series with -K2 coil sheet 08      |
| -F2    | 97-98 (NO) | PLC input X11 - OL tripped sheet 06   |
| -Q2    | aux 1 NO   | spare (future MCB trip alarm)<br>-    |
| -Q3    | aux 1 NO   | spare (future MCB trip alarm)<br>-    |

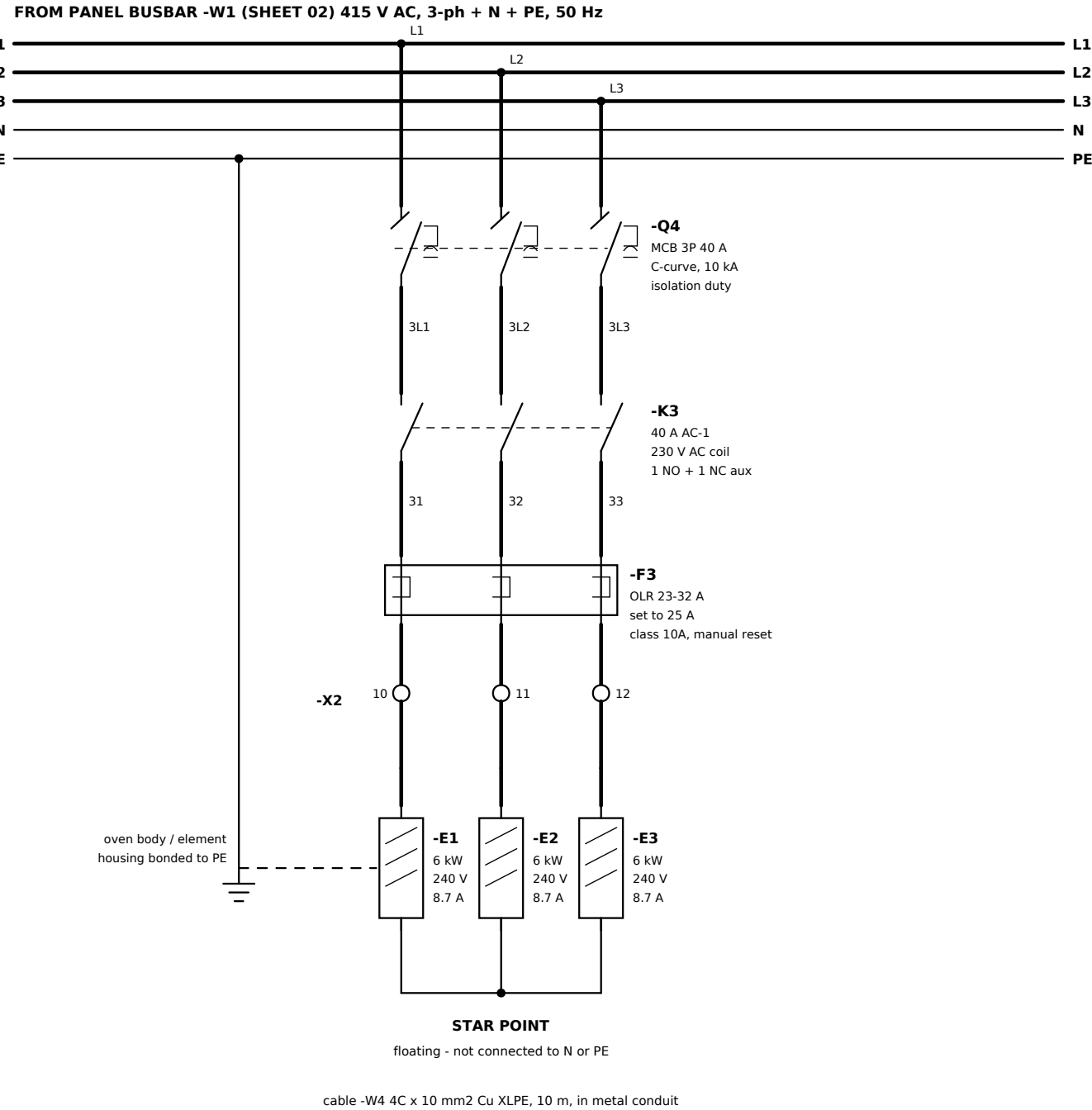
NOTES:

- Q2 / -Q3 are motor-protection circuit breakers providing short-circuit and isolation duty; the thermal overload relays -F1 / -F2 provide the running overload and phase-loss protection and give the trip signal to the PLC.
- Set both MPCB and OLR to the motor nameplate FLC after commissioning - the values shown assume 3.5 A.
- Overload relays are set to MANUAL RESET. A tripped overload must be reset at the panel; the PLC will not restart the blower until the trip input clears and the corresponding fault is acknowledged on the HMI.
- Both starters are direct-on-line. Motor direction to be checked at commissioning - airflow must be towards the oven plenum. Swap two phases at -X2 if rotation is incorrect.
- Blower contactors -K1 / -K2 are energised only when the PLC output Y0 / Y1 is ON - see sheets 07 and 08.

|                                                        |        |       |            |                        |         |     |
|--------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER      |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>POWER CIRCUIT - BLOWER MOTORS -M1 / -M2 |        |       |            |                        |         |     |
| DRAWN                                                  | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                | -      | SCALE | NTS        | OVN-2026-01            | 03 / 13 | 3   |

POWER CIRCUIT - HEATER BANK -E1 / -E2 / -E3

3 x 6 kW resistive elements, star connected, 18 kW total, 25 A per phase



HEATER CIRCUIT CALCULATION

|                            |                                                                                    |
|----------------------------|------------------------------------------------------------------------------------|
| Total connected load       | $P = 3 \times 6 \text{ kW} = 18 \text{ kW}$                                        |
| Line current (star, 415 V) | $I = 18000 / (1.732 \times 415) = 25.0 \text{ A}$                                  |
| Element voltage (star)     | $U = 415 / 1.732 = 240 \text{ V}$                                                  |
| Element current            | $I = 6000 / 240 = 25.0 \text{ A per element}$                                      |
| Element resistance (cold)  | $R = 240 / 25 = 9.6 \text{ ohm nominal}$                                           |
| Cable sizing (25 A, 40 C)  | 10 mm2 Cu -> 57 A capacity, derated 0.87<br><br>$= 49 \text{ A} > 25 \text{ A OK}$ |
| MCB -Q4                    | 40 A C-curve $> 1.45 \times 25 \text{ A} = 36 \text{ A OK}$                        |
| Overload -F3               | set 25 A, class 10A                                                                |
| Volt drop at 10 m          | approx. 0.7 V (0.17 %) OK                                                          |

CONTACTOR -K3 DUTY (see NOTE 6)

-K3 is switched by PLC output Y2, which follows the PID controller demand contact on input X6.

With a PID cycle time of 20 s the contactor would make and break up to 180 operations per hour, i.e. about 1.5 million operations per year on a 2-shift pattern.

The PLC program therefore imposes a minimum ON time and a minimum OFF time of 20 s each on Y2 (anti-chatter timers T\_HTR\_MIN\_ON / T\_HTR\_MIN\_OFF), which caps the switching rate at 90 operations per hour.

Set the controller cycle time to 20 s or longer to suit.

HEATER PROTECTION - LAYERS

- 1
- Q4 MCB 3P 40 A**  
short-circuit protection and isolation of the heater feeder
- 2
- F3 thermal overload relay**  
overload, element failure and phase asymmetry; manual reset
- 3
- B3 high-limit thermostat**  
independent over-temperature trip, HARD-WIRED in the -K3 coil circuit (sheet 08) - it drops the heater even if the PLC or the PID controller fails. Manual reset at the panel.
- 4
- A2 PID controller**  
closed-loop process temperature control; sensor-break alarm to X7
- 5
- PLC -A1**  
start permissives, heat-up watchdog alarm and manual-test interlocks

IMPORTANT:

The door interlock does NOT break the heater circuit. The heater stays energised while the oven door is open, as required by the process. Layer 3 is then the only protection limiting the oven temperature, so -B3 must be function-tested at every planned maintenance.

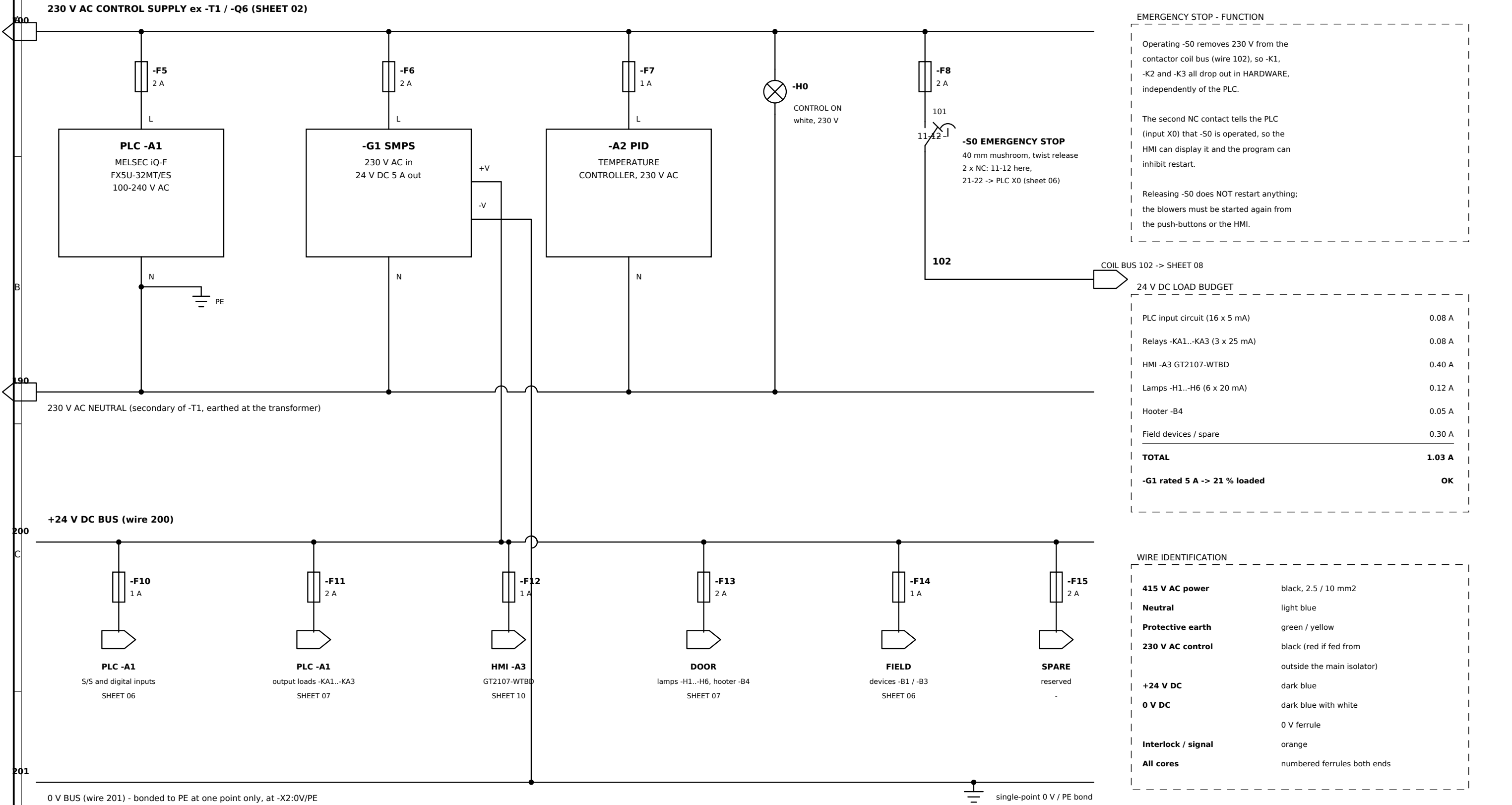
NOTES:

- Elements are connected in STAR with the star point floating. Do not connect the star point to the system neutral - a single element failure would then unbalance the neutral current.
- Q4 is C-curve. The heater is a purely resistive load with a cold-start inrush of about 1.3 x In for a few seconds.
- Element terminal box on the oven is to be wired with high-temperature silicone/glass-fibre insulated cable, rated 180 C minimum, from the gland plate to the element studs.
- Insulation resistance of the element bank to be measured before energising: > 1 Mohm at 500 V DC when cold.
- F3 also protects against the loss of one element (current asymmetry) - set it to the measured balanced current.
- Contactor -K3 is an AC-1 rated device switching a resistive load. See the duty block above: the PLC limits the switching frequency so that the mechanical and electrical life of -K3 is not consumed prematurely.

|                                                            |        |       |            |                        |         |     |
|------------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER          |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>POWER CIRCUIT - HEATER BANK -E1 / -E2 / -E3 |        |       |            |                        |         |     |
| DRAWN                                                      | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                    | -      | SCALE | NTS        | OVN-2026-01            | 04 / 13 | 3   |

CONTROL SUPPLY, EMERGENCY STOP & 24 V DC DISTRIBUTION

230 V AC control circuit ex -T1 / 24 V DC ex -G1

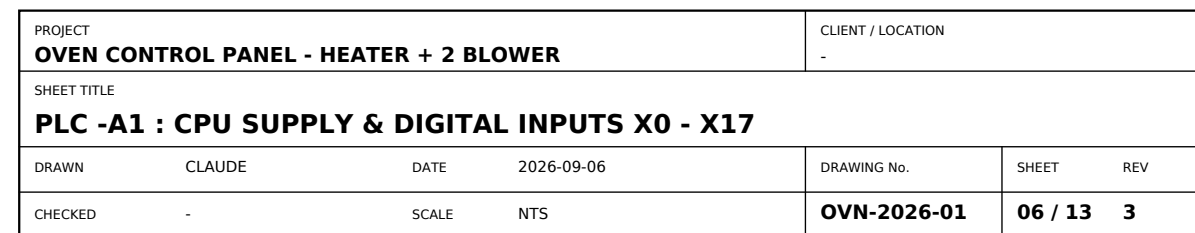
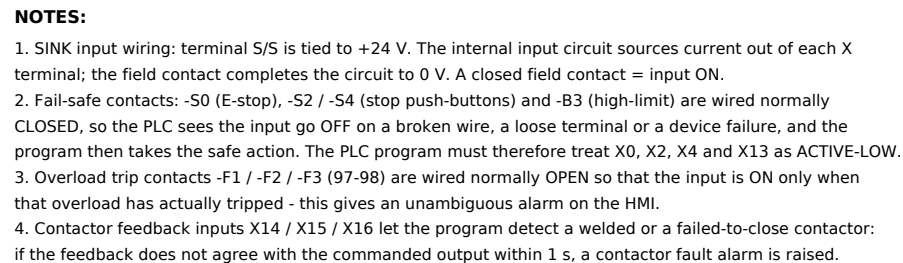


NOTES:

1. -F5 to -F15 are DIN-rail fuse holders with 5 x 20 mm cartridge fuses; the rating is marked beside each holder.
2. The PLC, the SMPS and the PID controller are fed UPSTREAM of the emergency stop so that they stay alive during an E-stop; only the contactor coil bus (wire 102) is interrupted, and that is what removes power from the loads.
3. Wire crossings drawn with a semicircular hop are NOT connected. A junction is shown by a solid dot.

|                                                       |        |       |            |                   |         |     |
|-------------------------------------------------------|--------|-------|------------|-------------------|---------|-----|
| PROJECT                                               |        |       |            | CLIENT / LOCATION |         |     |
| OVEN CONTROL PANEL - HEATER + 2 BLOWER                |        |       |            | -                 |         |     |
| SHEET TITLE                                           |        |       |            |                   |         |     |
| CONTROL SUPPLY, EMERGENCY STOP & 24 V DC DISTRIBUTION |        |       |            |                   |         |     |
| DRAWN                                                 | CLAUDE | DATE  | 2026-09-06 | DRAWING No.       | SHEET   | REV |
| CHECKED                                               | -      | SCALE | NTS        | OVN-2026-01       | 05 / 13 | 3   |

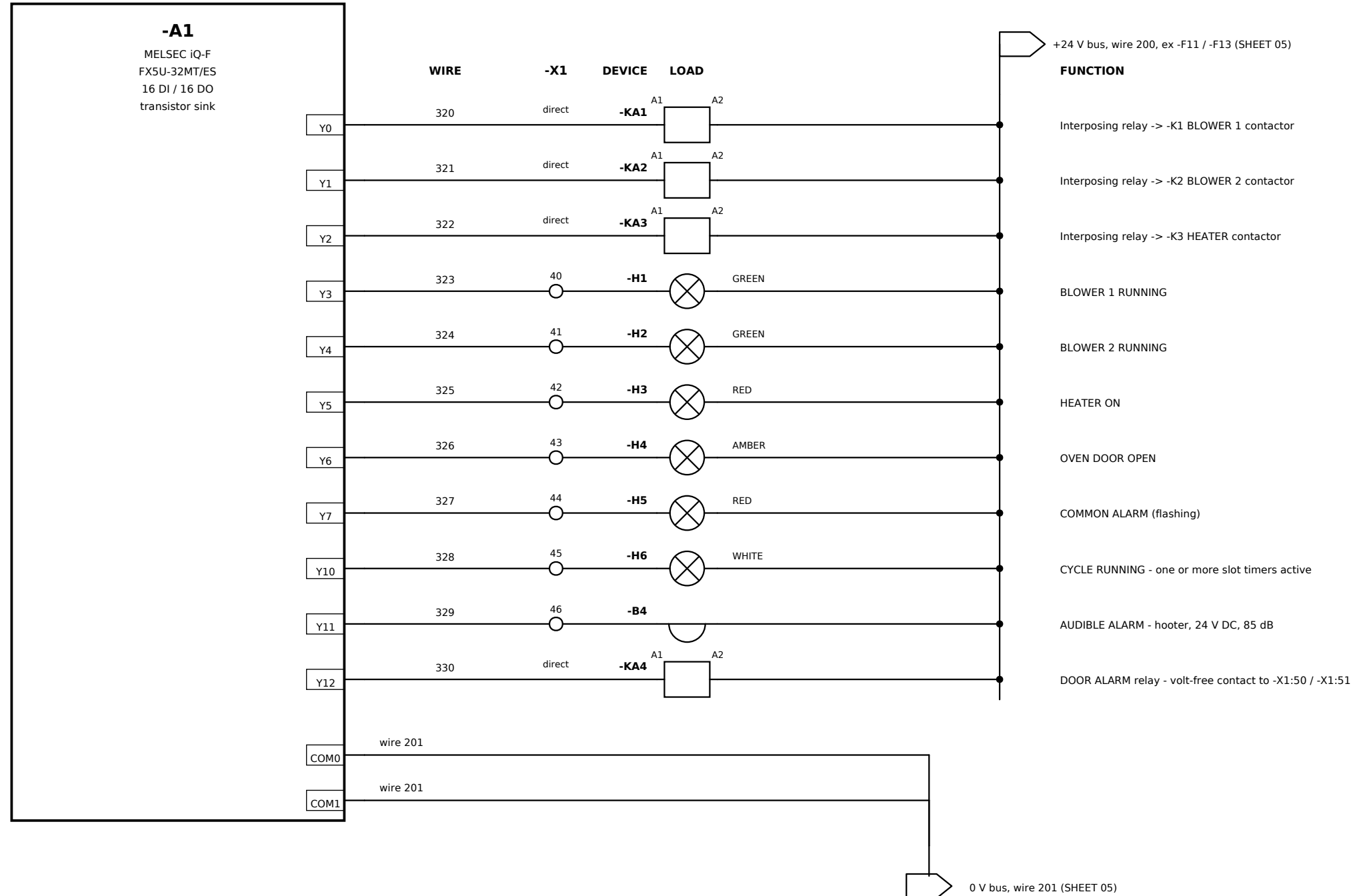
MELSEC iQ-F FX5U-32MT/ES - sink input wiring (S/S = +24 V)





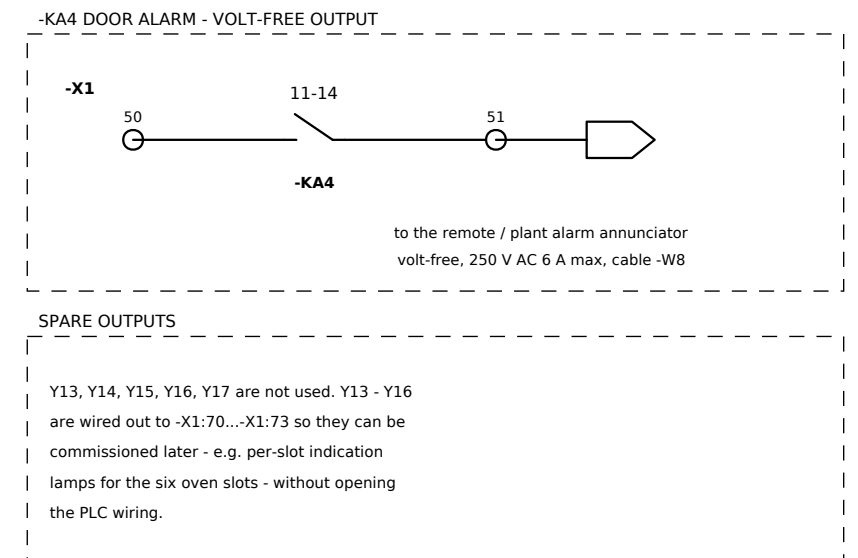
### PLC -A1 : DIGITAL OUTPUTS Y0 - Y17

MELSEC iQ-F FX5U-32MT/ES - transistor SINK outputs, 24 V DC, 0.5 A per point



**NOTES:**

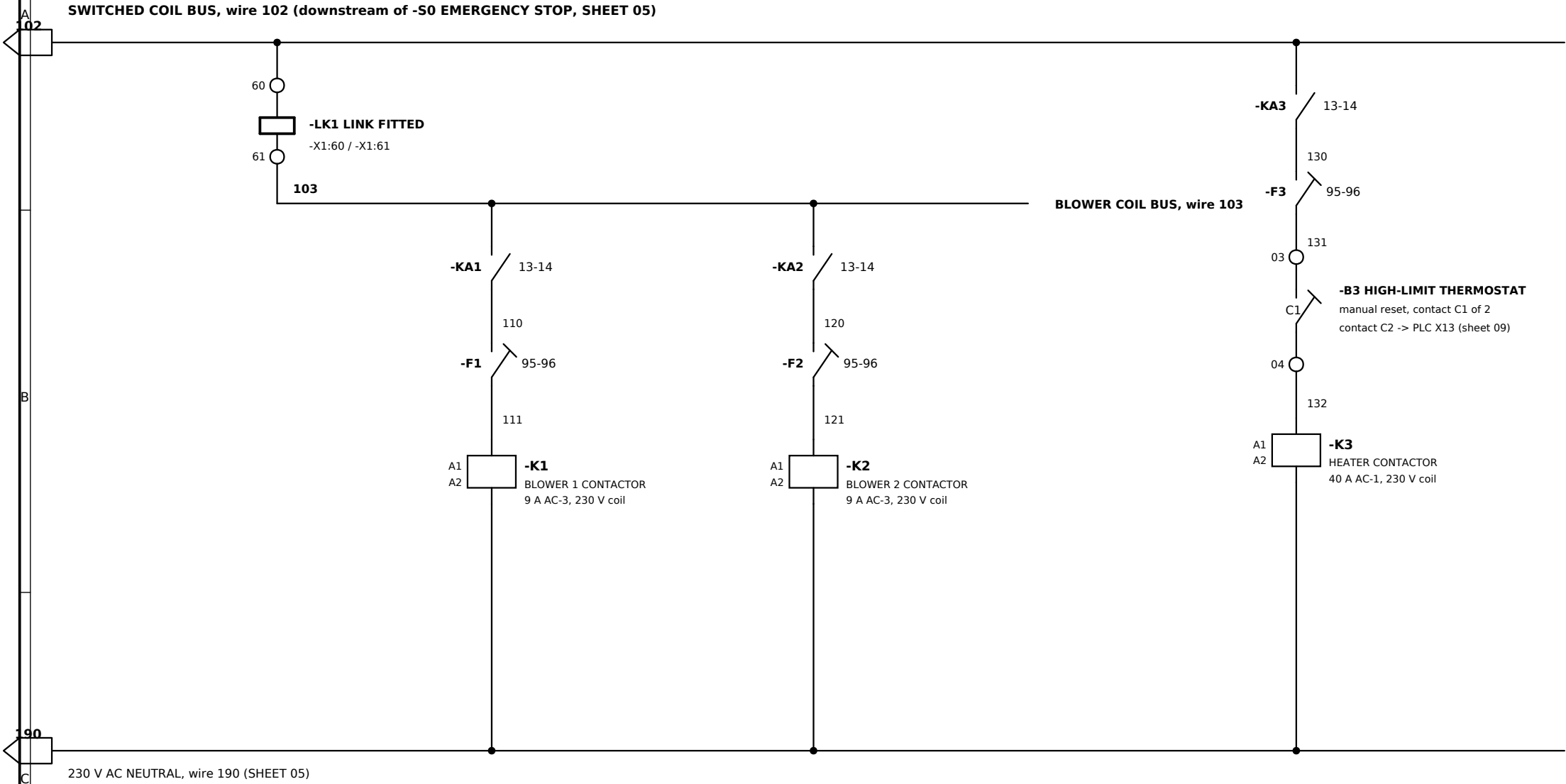
1. Outputs are transistor SINK type. The load is connected between +24 V (wire 200) and the Y terminal; the output transistor completes the circuit to 0 V through COM0 / COM1. Never connect a load between a Y terminal and 0 V - it will be permanently energised.
2. Each output point is rated 0.5 A; each common group is rated 0.8 A total. The loads used here (relay coils 25 mA, LED lamps 20 mA, hooter 50 mA) are well inside both limits.
3. Interposing relays -KA1 / -KA2 / -KA3 have an integral free-wheel diode fitted across the coil. Where a socket-mounted relay without a diode is used, fit a 1N4007 across A1 (+) / A2 (-) to protect the output.
4. Contactor coils are 230 V AC and are NOT driven directly from the PLC - see sheet 08.
5. -H5 COMMON ALARM is flashed by the program (1 s on / 1 s off) while an alarm is unacknowledged and is steady once accepted with -S5. The hooter -B4 silences on -S5 but the lamp stays on until the cause clears.
6. Y12 / -KA4 is the DOOR ALARM output. It is energised by either of the door-signal watchdog conditions - a chattering door switch, or no 'door closed' signal for 10 minutes - and gives a volt-free changeover contact on -X1:50 / -X1:51 for a remote annunciator. The program also flashes -H4 DOOR OPEN for this condition, so a faulty door switch is distinguishable at the panel from a door that is simply open.



|                                                   |  |                    |  |                        |  |              |  |
|---------------------------------------------------|--|--------------------|--|------------------------|--|--------------|--|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER |  |                    |  | CLIENT / LOCATION<br>- |  |              |  |
| SHEET TITLE<br>PLC -A1 : DIGITAL OUTPUTS Y0 - Y17 |  |                    |  |                        |  |              |  |
| DRAWN<br>CLAUDE                                   |  | DATE<br>2026-09-06 |  | DRAWING No.            |  | SHEET<br>REV |  |
| CHECKED<br>-                                      |  | SCALE<br>NTS       |  | OVN-2026-01            |  | 07 / 13 3    |  |

CONTACTOR COIL CIRCUITS & DOOR-INTERLOCK PROVISION

230 V AC coil circuits fed from the switched coil bus, wire 102



NOTES:

- Interposing relays -KA1 / -KA2 / -KA3 have 24 V DC coils driven from PLC outputs Y0 / Y1 / Y2 (sheet 07). Their volt-free 13-14 contacts switch the 230 V AC contactor coils shown here, so the PLC never carries 230 V and the coil circuit stays independent of the PLC output stage.
- Overload contacts -F1 / -F2 / -F3 (95-96) are in the coil circuits, so an overload trip drops its contactor in HARDWARE. The 97-98 contact of the same relay tells the PLC which overload tripped (sheet 06).
- B3 contact C1 is in the -K3 coil circuit only. An over-temperature therefore removes the heater without any involvement of the PLC or the PID controller. Contact C2 reports the same event to PLC input X13.
- Contactor auxiliary contacts 13-14 of -K1 / -K2 / -K3 report back to PLC inputs X14 / X15 / X16 so the program can detect a welded contactor or one that failed to pull in.
- Fit an RC suppressor across the coil of -K3 to limit the switching transient seen by -KA3.

HOW EACH COIL IS ENERGISED

- K1 / -K2 (blowers)**
- 102 present (E-stop released)
  - AND -LK1 fitted (or the future door switch closed)
  - AND PLC output Y0 / Y1 ON
  - AND that overload not tripped
- K3 (heater)**
- 102 present (E-stop released)
  - AND PLC output Y2 ON
  - AND -F3 not tripped
  - AND -B3 high-limit not tripped
- The door interlock acts on Y0 / Y1 in the PLC program only - it is NOT in the -K3 chain, so the heater stays ON while the door is open.

-LK1 UPGRADE TO A HARD-WIRED INTERLOCK

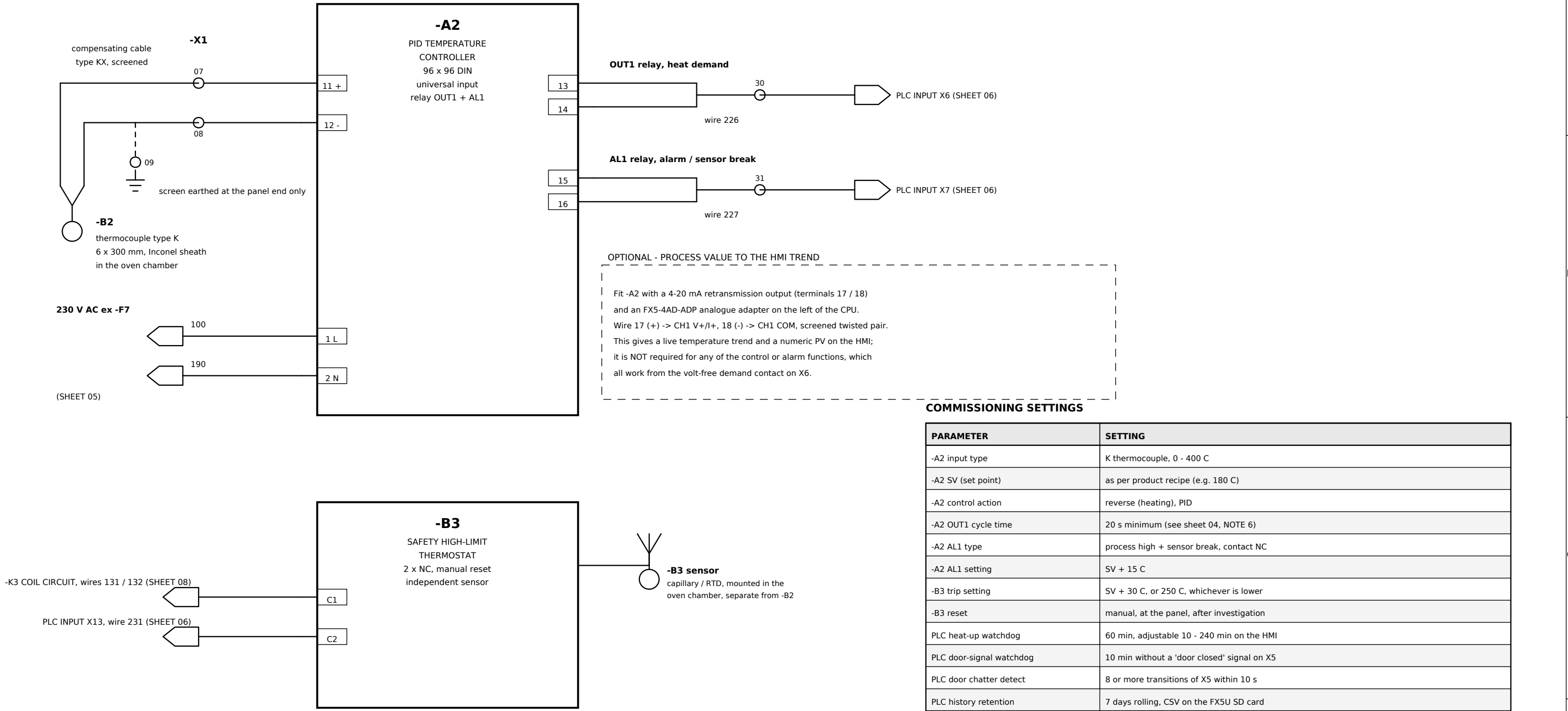
- As instructed, the door switch -B1 goes to PLC input X5 only and the blowers are stopped by the program. The wire link -LK1 between -X1:60 and -X1:61 keeps the blower coil bus permanently made.
- To convert to a hard-wired interlock later:
- Remove the link -LK1.
  - Land the NC contact of a second (safety-rated) door switch -B1.2 on -X1:60 and -X1:61.
  - No other panel change is needed; X5 keeps working for the counters and the HMI status.

|                                                                   |        |       |            |                        |         |     |
|-------------------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER                 |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>CONTACTOR COIL CIRCUITS & DOOR-INTERLOCK PROVISION |        |       |            |                        |         |     |
| DRAWN                                                             | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                           | -      | SCALE | NTS        | OVN-2026-01            | 08 / 13 | 3   |



TEMPERATURE CONTROLLER, THERMOCOUPLE & OVER-TEMPERATURE PROTECTION

-A2 PID controller, -B2 type K thermocouple, -B3 independent high-limit thermostat



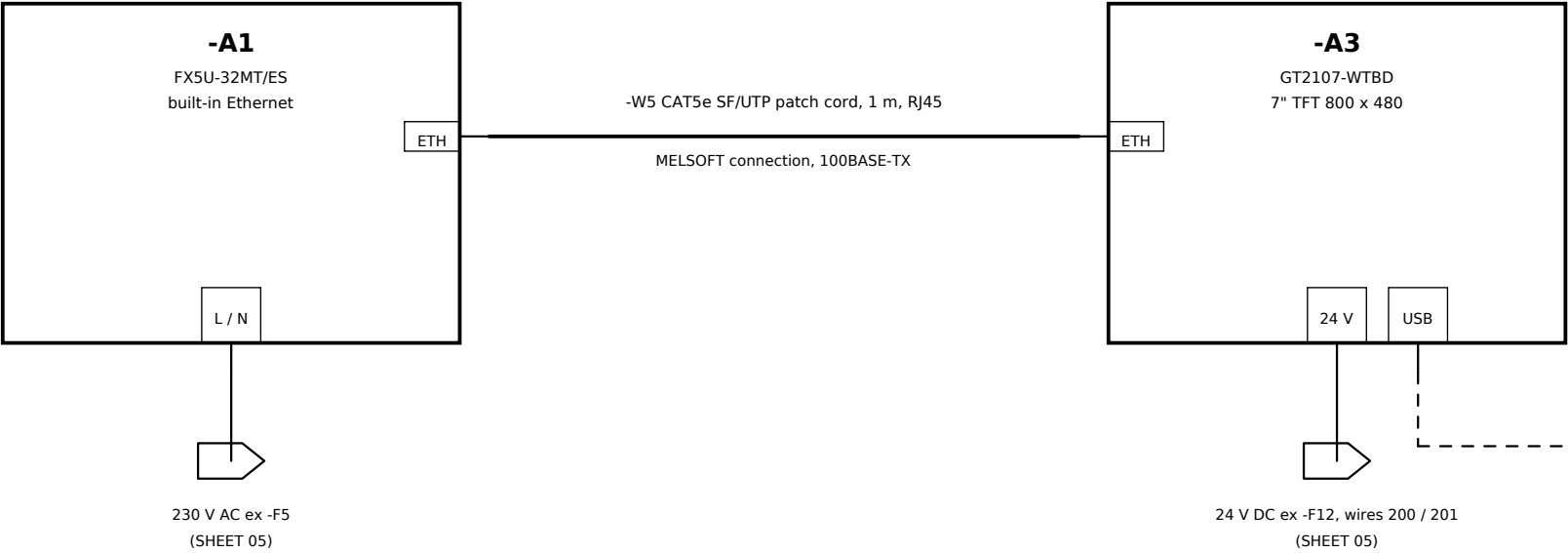
NOTES:

- A2 keeps full local control of the process temperature. The PLC does not compute the temperature loop; it simply repeats the controller's demand contact to the heater contactor and adds the permissives and alarms.
- The demand contact on X6 is volt-free. If the controller supplied is an SSR-drive (12 V DC pulse) type instead of a relay type, fit an interposing relay -KA5 (12 V DC coil) and take its NO contact to X6 - the PLC program is unchanged.
- B2 and the -B3 sensor must be separate sensing elements in separate pockets. A single shared sensor would defeat the whole purpose of the independent over-temperature layer.
- Heat-up watchdog: the PLC starts a timer when the demand contact X6 closes. If X6 stays closed continuously for longer than the watchdog time - i.e. the controller never gets to set point and drops out - the alarm 'FAILURE TO REACH SETPOINT' is raised on the HMI and the hooter sounds. The heater is not tripped by this alarm; it is an operator warning, since the real protection against over-temperature is -B3.

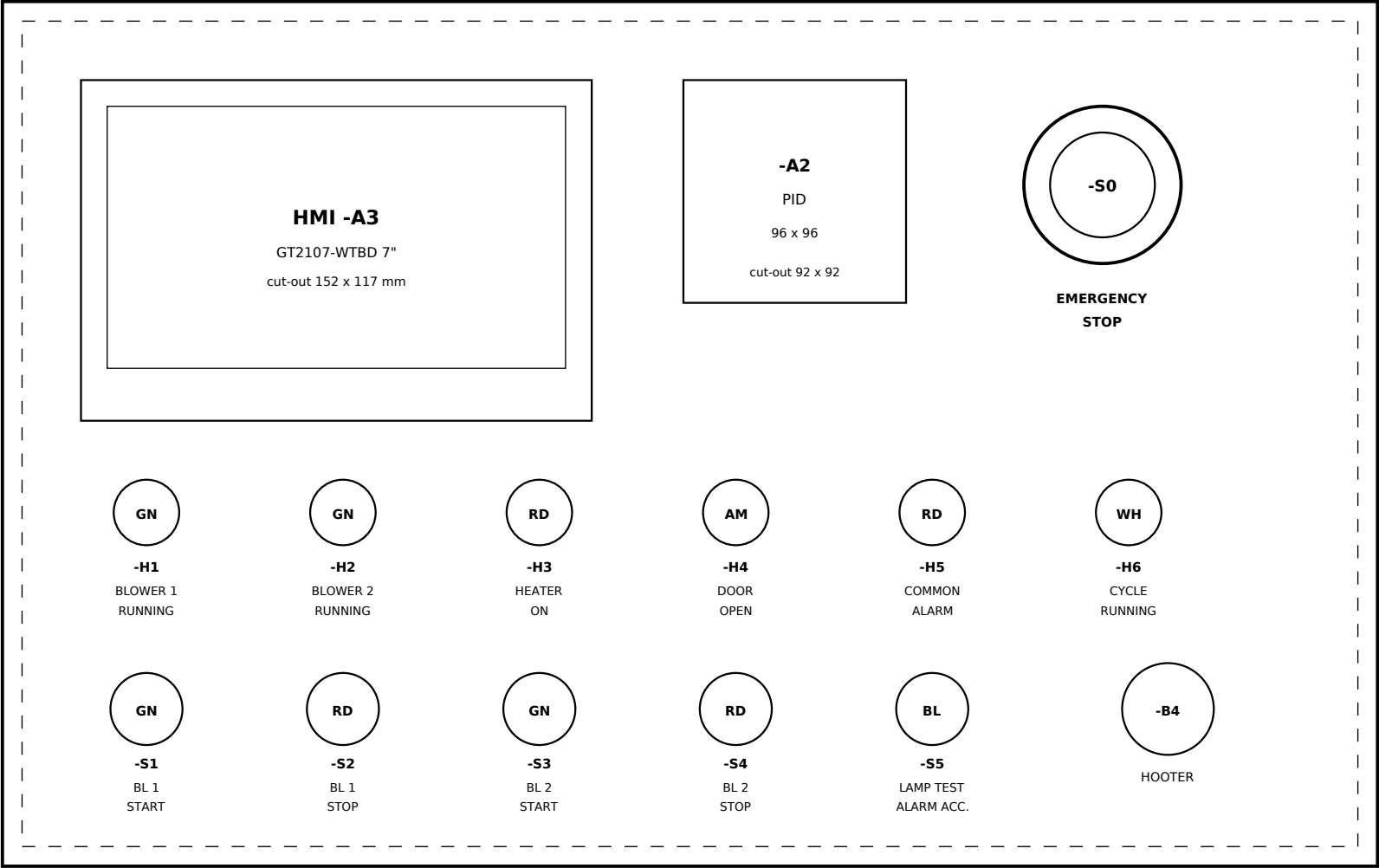
|                                                                                   |        |       |            |                        |         |     |
|-----------------------------------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER                                 |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>TEMPERATURE CONTROLLER, THERMOCOUPLE & OVER-TEMPERATURE PROTECTION |        |       |            |                        |         |     |
| DRAWN                                                                             | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                                           | -      | SCALE | NTS        | OVN-2026-01            | 09 / 13 | 3   |

HMI -A3, ETHERNET NETWORK & PANEL DOOR ARRANGEMENT

GOT2000 GT2107-WTBD <-> MELSEC iQ-F FX5U-32MT/ES, 100BASE-TX



PANEL DOOR - EXTERNAL ARRANGEMENT (view on front, 800 x 600 mm door)



NETWORK ADDRESSING

| DEVICE         | IP ADDRESS    | NOTES                    |
|----------------|---------------|--------------------------|
| -A1 FX5U       | 192.168.3.250 | built-in port, default   |
| -A3 GT2107     | 192.168.3.18  | GOT default              |
| Engineering PC | 192.168.3.10  | GX Works3 / GT Designer3 |
| Subnet mask    | 255.255.255.0 | -                        |
| Gateway        | not used      | isolated machine network |

HMI SCREEN STRUCTURE (see the HMI project for detail)

| SCREEN | TITLE                           | ROLE    |
|--------|---------------------------------|---------|
| B-1000 | HOME / OVERVIEW                 | -       |
| B-1100 | SLOT TIMERS 1-6                 | -       |
| B-1200 | COUNTERS - door, lots           | -       |
| B-1300 | ALARMS - live & history         | -       |
| B-1400 | MANUAL TEST - blowers, heater   | MAINT   |
| B-1500 | SETTINGS - watchdog, presets    | MAINT   |
| B-1600 | LOGIN - passcode keypad         | -       |
| B-1700 | TIMER EARLY RESET               | MAINT   |
| B-1800 | COUNTER RESET - door and lots   | QUALITY |
| B-1900 | HISTORY - log view & USB export | -       |

'-' = operator, no login. MAINT and QUALITY are two SEPARATE passcodes - neither inherits the other's rights (sheet 01, NOTE 11).

NOTES:

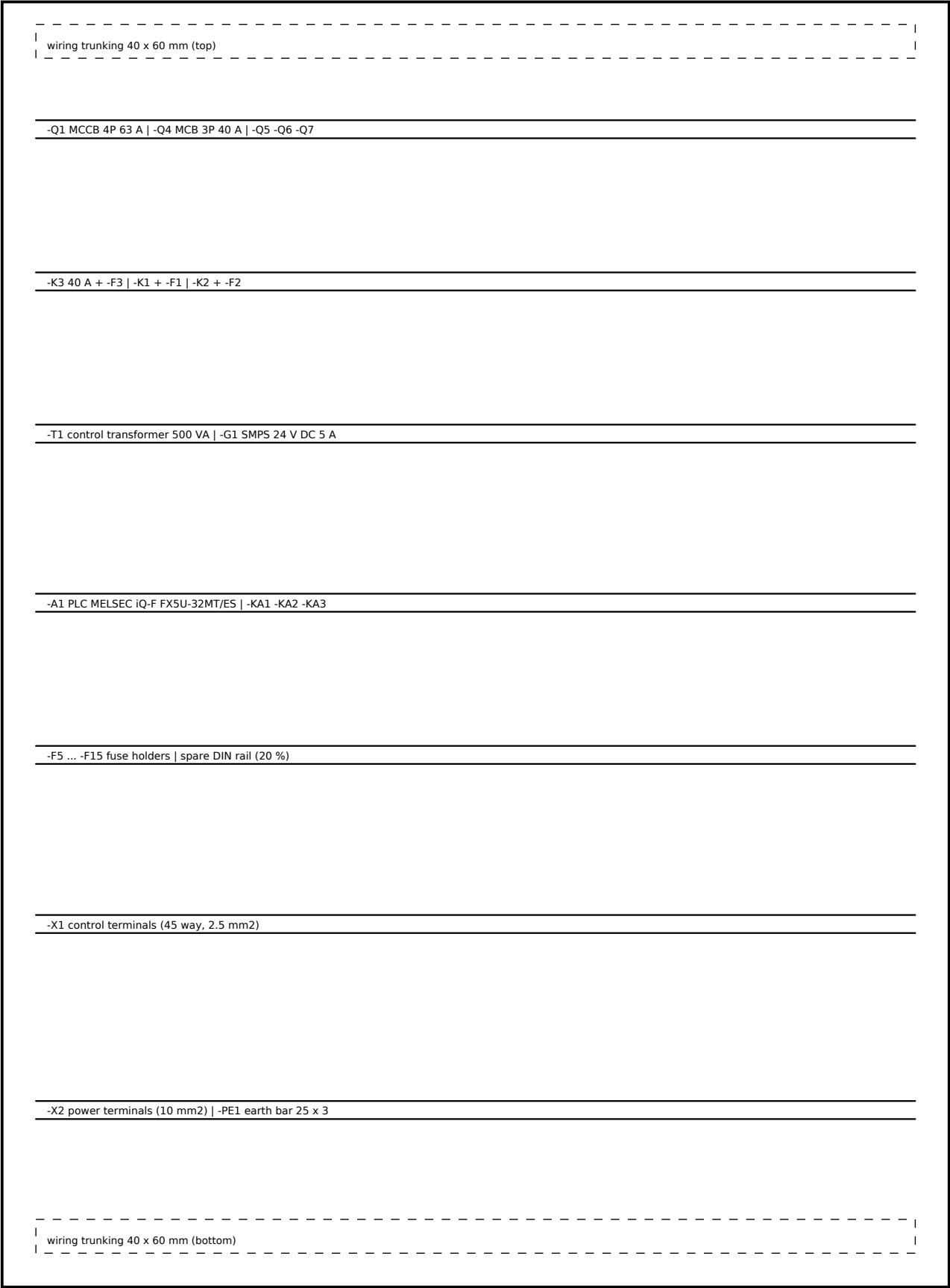
- The HMI is powered from -F12 and is NOT interrupted by the emergency stop, so alarms and the timer values stay visible.
- Slot timer values, counters and the alarm history are held in PLC latched memory (see the PLC program device list), not in the HMI, so they survive a power failure and an HMI replacement.
- The GOT battery (GT11-50BAT) is fitted for the real-time clock used by the alarm and the operation-log time stamps.
- The 7-day alarm and event history is written as CSV to the SD card in the FX5U, using the CPU data-logging function, and is exported to a USB stick from screen B-1900.
- Door cut-outs to be made before painting; use the gaskets supplied to keep IP54.

|                                                                   |        |       |            |                        |         |     |
|-------------------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER                 |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>HMI -A3, ETHERNET NETWORK & PANEL DOOR ARRANGEMENT |        |       |            |                        |         |     |
| DRAWN                                                             | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                           | -      | SCALE | NTS        | OVN-2026-01            | 10 / 13 | 3   |

PANEL GENERAL ARRANGEMENT & GLAND PLATE

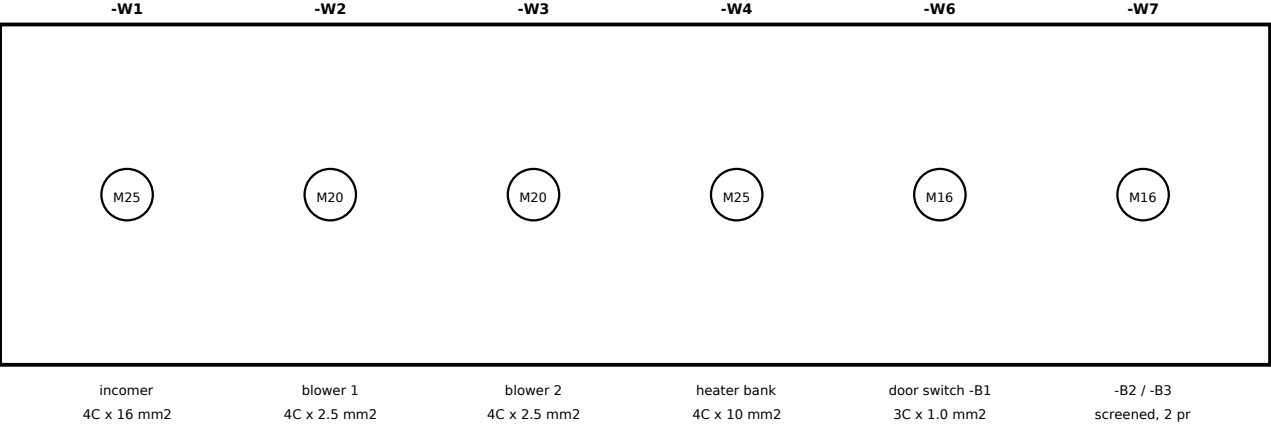
Wall-mounted enclosure 800 H x 600 W x 250 D mm, IP54, RAL 7035

MOUNTING PLATE - INTERNAL LAYOUT (scale approx. 1:3.3)



mounting plate 560 x 760 mm

GLAND PLATE - BOTTOM OF ENCLOSURE



ENCLOSURE & CONSTRUCTION

| ITEM            | SPECIFICATION                                                |
|-----------------|--------------------------------------------------------------|
| Enclosure       | 800 H x 600 W x 250 D mm, mild steel 1.5 mm, IP54            |
| Finish          | powder coated RAL 7035, plate RAL 9003                       |
| Mounting        | wall mounted, 4 x M8 bolts, bottom of panel 1000 mm AFFL     |
| Door            | single door, 3-point lock, earthed with 4 mm2 braid          |
| Ventilation     | filter fan 100 m3/h + exhaust filter, thermostat set 35 C    |
| Cable entry     | bottom gland plate, removable, glands per the detail above   |
| Internal wiring | in slotted trunking, minimum 60 % spare capacity             |
| Segregation     | 415 V power and 24 V DC control in separate trunking runs    |
| Earthing        | copper earth bar -PE1 25 x 3 mm, all doors and plates bonded |
| Labelling       | engraved traffolyte for devices, printed ferrules for cores  |
| Spare capacity  | 20 % free DIN rail and 4 spare terminals per strip           |
| Tests           | IR 500 V DC, flash 1.5 kV 1 min, functional test to sheet 13 |

INSTALLATION REQUIREMENTS

1. Keep 800 mm clear working space in front of the panel (IEC 60204-1).

2. The panel must not be mounted on the oven skin or within 1 m of it - radiated heat will drive the internal temperature above the PLC rating (55 C). Mount on the adjacent wall or on a free-standing frame.

3. Thermocouple and 4-20 mA cables to be run at least 300 mm away from the motor and heater power cables, crossing at right angles only.

4. The oven body, the element housing and the blower frames must all be bonded to -PE1. Measure the bond resistance: < 0.1 ohm.

NOTES:

1. Layout shown is indicative; the panel builder may rearrange within the same enclosure provided the segregation and clearance requirements are kept.

2. Fit an internal panel light and a 6 A socket outlet fed from -Q7 (RCBO).

3. Fit a door-mounted pocket for this drawing set and the O&M manual.

|                                                        |        |       |            |                        |         |     |
|--------------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER      |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>PANEL GENERAL ARRANGEMENT & GLAND PLATE |        |       |            |                        |         |     |
| DRAWN                                                  | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                                | -      | SCALE | NTS        | OVN-2026-01            | 11 / 13 | 3   |

12345678

# TERMINAL STRIP SCHEDULE -X1 / -X2

All field and door-loom wiring terminates here; panel-plate devices are wired direct

A

B

C

D

12345678

## -X1 CONTROL TERMINALS (2.5 mm2)

| TERM | WIRE | DEVICE   | DESCRIPTION                                  | SH. |
|------|------|----------|----------------------------------------------|-----|
| 01   | 225  | -B1:13   | Oven door position switch - contact 13       | 06  |
| 02   | 201  | -B1:14   | Oven door position switch - contact 14 (0 V) | 06  |
| 03   | 102  | -B3:C1   | High-limit thermostat contact 1 - from -KA3  | 08  |
| 04   | 132  | -B3:C1   | High-limit thermostat contact 1 - to -K3 A1  | 08  |
| 05   | 231  | -B3:C2   | High-limit thermostat contact 2 - to PLC X13 | 06  |
| 06   | 201  | -B3:C2   | High-limit thermostat contact 2 - 0 V return | 06  |
| 07   | TC+  | -B2:+    | Thermocouple type K, positive (green)        | 09  |
| 08   | TC-  | -B2:-    | Thermocouple type K, negative (white)        | 09  |
| 09   | SCR  | -B2 scr  | Thermocouple screen - earthed at panel only  | 09  |
| 10   | PE   | oven     | Oven body bonding conductor                  | 04  |
| 20   | 220  | -S0:21   | Emergency stop, contact 21-22 -> PLC X0      | 06  |
| 21   | 221  | -S1:13   | Blower 1 START push-button -> PLC X1         | 06  |
| 22   | 222  | -S2:11   | Blower 1 STOP push-button -> PLC X2          | 06  |
| 23   | 223  | -S3:13   | Blower 2 START push-button -> PLC X3         | 06  |
| 24   | 224  | -S4:11   | Blower 2 STOP push-button -> PLC X4          | 06  |
| 30   | 226  | -A2:OUT1 | PID controller heat demand -> PLC X6         | 09  |
| 31   | 227  | -A2:AL1  | PID controller alarm -> PLC X7               | 09  |
| 33   | 235  | -S5:13   | Lamp test / alarm accept -> PLC X17          | 06  |
| 34   | 201  | 0 V      | 0 V common for door-mounted push-buttons     | 06  |

## -X2 POWER TERMINALS (10 mm2 + PE)

| TERM | WIRE | DEVICE | DESCRIPTION                                       |
|------|------|--------|---------------------------------------------------|
| 01   | 11   | -M1 U1 | Blower 1 motor, phase U                           |
| 02   | 12   | -M1 V1 | Blower 1 motor, phase V                           |
| 03   | 13   | -M1 W1 | Blower 1 motor, phase W                           |
| 04   | 21   | -M2 U1 | Blower 2 motor, phase U                           |
| 05   | 22   | -M2 V1 | Blower 2 motor, phase V                           |
| 06   | 23   | -M2 W1 | Blower 2 motor, phase W                           |
| 10   | 31   | -E1    | Heater element 1, line                            |
| 11   | 32   | -E2    | Heater element 2, line                            |
| 12   | 33   | -E3    | Heater element 3, line                            |
| PE   | PE   | PE bar | Protective earth - motors, oven body, gland plate |

## -X1 CONTROL TERMINALS (continued)

| TERM | WIRE | DEVICE  | DESCRIPTION                                             | SH. |
|------|------|---------|---------------------------------------------------------|-----|
| 35   | 201  | 0 V     | 0 V common for door-mounted push-buttons                | 06  |
| 40   | 323  | -H1     | Lamp BLOWER 1 RUNNING (green) <- Y3                     | 07  |
| 41   | 324  | -H2     | Lamp BLOWER 2 RUNNING (green) <- Y4                     | 07  |
| 42   | 325  | -H3     | Lamp HEATER ON (red) <- Y5                              | 07  |
| 43   | 326  | -H4     | Lamp DOOR OPEN (amber) <- Y6                            | 07  |
| 44   | 327  | -H5     | Lamp COMMON ALARM (red) <- Y7                           | 07  |
| 45   | 328  | -H6     | Lamp CYCLE RUNNING (white) <- Y10                       | 07  |
| 46   | 329  | -B4     | Hooter 24 V DC <- Y11                                   | 07  |
| 47   | 200  | +24 V   | +24 V common for door-mounted lamps / hooter            | 07  |
| 48   | 200  | +24 V   | +24 V common for door-mounted lamps / hooter            | 07  |
| 50   | VF1  | -KA4:11 | DOOR ALARM volt-free contact - remote / plant alarm     | 07  |
| 51   | VF2  | -KA4:14 | DOOR ALARM volt-free contact - remote / plant alarm     | 07  |
| 60   | 102  | LINK    | PROVISION - hard-wired door interlock, link -LK1 fitted | 08  |
| 61   | 103  | LINK    | PROVISION - hard-wired door interlock, link -LK1 fitted | 08  |
| 70   | -    | -       | SPARE                                                   | -   |
| 71   | -    | -       | SPARE                                                   | -   |
| 72   | -    | -       | SPARE                                                   | -   |
| 73   | -    | -       | SPARE                                                   | -   |

### TERMINAL STRIP CONVENTIONS

1. -X1 is a single strip in terminal-number order with a marked end plate at each end.
2. Terminals 01-10 are the OVEN field devices, 20-35 the panel DOOR loom, 40-48 the door indication loom, 60-61 the door-interlock provision and 70-73 spare.
3. Every core carries a printed ferrule with the wire number shown, at BOTH ends.
4. A grey terminal is used for all signals, light blue for 0 V and green/yellow for PE.
5. No more than TWO conductors in any one terminal; use a jumper comb for commons.
6. -X1:60 / -X1:61 carry the link -LK1. Do not remove the link unless a hard-wired door interlock is being fitted - the blowers will not run with the link removed.
7. Spare terminals 70-73 are pre-wired to PLC outputs Y12-Y15 for future use.

### NOTES:

1. Wire numbers follow the scheme: 1xx = 230 V AC control, 2xx = 24 V DC inputs, 3xx = 24 V DC outputs, and power cores are numbered by feeder (1L1, 2L1, 31 ...).
2. Check every ferrule against this schedule at pre-delivery inspection and again after installation; a mis-numbered core is the single most common cause of a failed FAT.

|                                                   |        |       |            |                        |         |     |
|---------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>TERMINAL STRIP SCHEDULE -X1 / -X2  |        |       |            |                        |         |     |
| DRAWN                                             | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                           | -      | SCALE | NTS        | OVN-2026-01            | 12 / 13 | 3   |

I/O SCHEDULE & BILL OF MATERIALS

MELSEC iQ-F FX5U-32MT/ES - 16 digital inputs, 16 digital outputs

DIGITAL INPUTS

| ADDR | WIRE | -X1 | DEVICE | CONTACT | DESCRIPTION                                         |
|------|------|-----|--------|---------|-----------------------------------------------------|
| X0   | 220  | 20  | -S0    | 21-22   | EMERGENCY STOP - healthy (NC, fail-safe)            |
| X1   | 221  | 21  | -S1    | 13-14   | BLOWER 1 START push-button                          |
| X2   | 222  | 22  | -S2    | 11-12   | BLOWER 1 STOP push-button (NC, fail-safe)           |
| X3   | 223  | 23  | -S3    | 13-14   | BLOWER 2 START push-button                          |
| X4   | 224  | 24  | -S4    | 11-12   | BLOWER 2 STOP push-button (NC, fail-safe)           |
| X5   | 225  | 01  | -B1    | 13-14   | OVEN DOOR CLOSED - position switch                  |
| X6   | 226  | 30  | -A2    | OUT1    | TEMP CONTROLLER heat demand (volt-free)             |
| X7   | 227  | 31  | -A2    | AL1     | TEMP CONTROLLER alarm / sensor break (NC = healthy) |
| X10  | 228  | -   | -F1    | 97-98   | BLOWER 1 overload TRIPPED                           |
| X11  | 229  | -   | -F2    | 97-98   | BLOWER 2 overload TRIPPED                           |
| X12  | 230  | -   | -F3    | 97-98   | HEATER overload TRIPPED                             |
| X13  | 231  | 05  | -B3    | C2      | HIGH-LIMIT THERMOSTAT healthy (NC = healthy)        |
| X14  | 232  | -   | -K1    | 13-14   | BLOWER 1 contactor closed - feedback                |
| X15  | 233  | -   | -K2    | 13-14   | BLOWER 2 contactor closed - feedback                |
| X16  | 234  | -   | -K3    | 13-14   | HEATER contactor closed - feedback                  |
| X17  | 235  | 33  | -S5    | 13-14   | LAMP TEST / ALARM ACCEPT push-button                |

DIGITAL OUTPUTS

| ADDR | WIRE | -X1 | DEVICE | COLOUR | DESCRIPTION                                             |
|------|------|-----|--------|--------|---------------------------------------------------------|
| Y0   | 320  | -   | -KA1   | -      | Interposing relay -> -K1 BLOWER 1 contactor             |
| Y1   | 321  | -   | -KA2   | -      | Interposing relay -> -K2 BLOWER 2 contactor             |
| Y2   | 322  | -   | -KA3   | -      | Interposing relay -> -K3 HEATER contactor               |
| Y3   | 323  | 40  | -H1    | GREEN  | BLOWER 1 RUNNING                                        |
| Y4   | 324  | 41  | -H2    | GREEN  | BLOWER 2 RUNNING                                        |
| Y5   | 325  | 42  | -H3    | RED    | HEATER ON                                               |
| Y6   | 326  | 43  | -H4    | AMBER  | OVEN DOOR OPEN                                          |
| Y7   | 327  | 44  | -H5    | RED    | COMMON ALARM (flashing)                                 |
| Y10  | 328  | 45  | -H6    | WHITE  | CYCLE RUNNING - one or more slot timers active          |
| Y11  | 329  | 46  | -B4    | -      | AUDIBLE ALARM - hooter, 24 V DC, 85 dB                  |
| Y12  | 330  | -   | -KA4   | -      | DOOR ALARM relay - volt-free contact to -X1:50 / -X1:51 |
| Y13  | -    | -   | -      | -      | SPARE                                                   |
| Y14  | -    | -   | -      | -      | SPARE                                                   |
| Y15  | -    | -   | -      | -      | SPARE                                                   |
| Y16  | -    | -   | -      | -      | SPARE                                                   |
| Y17  | -    | -   | -      | -      | SPARE                                                   |

NOTES:

1. I/O count used: 16 of 16 digital inputs (0 spare), 11 of 16 digital outputs (5 spare).
- The input card is full. If any further input is needed - for example a second door switch, a per-slot load sensor or an airflow proving switch - add an FX5-16EX/ES input extension block on the right of the CPU; no rewiring of the existing inputs is needed.
2. Quantities in the bill of materials exclude consumables (ferrules, cable ties, glands, labels, wire).
3. Equivalent devices from another manufacturer may be substituted provided the ratings, the utilisation category and the terminal arrangement are the same, and the substitution is recorded on the as-built.
4. All settings marked 'set to ...' on sheets 03, 04 and 09 must be recorded on the commissioning sheet.

BILL OF MATERIALS

| #  | TAG        | DESCRIPTION                                               | TYPE / MAKE                      | QTY |
|----|------------|-----------------------------------------------------------|----------------------------------|-----|
| 1  | -Q1        | MCCB 4P 63 A, Icu 25 kA, adj. 40-63 A                     | Mitsubishi NF63-CV 4P 63A        | 1   |
| 2  | -Q2/-Q3    | Motor protection circuit breaker 3P 2.5-4 A               | Schneider GV2ME08                | 2   |
| 3  | -Q4        | MCB 3P 40 A C-curve 10 kA                                 | Mitsubishi BH-DN 3P C40          | 1   |
| 4  | -Q5        | MCB 2P 6 A C-curve                                        | Mitsubishi BH-DN 2P C6           | 1   |
| 5  | -Q6        | MCB 1P 4 A C-curve                                        | Mitsubishi BH-DN 1P C4           | 1   |
| 6  | -Q7        | RCBO 2P 6 A, 30 mA                                        | Mitsubishi BV-DN 2P C6           | 1   |
| 7  | -K1/-K2    | Contactors 9 A AC-3, 230 V AC coil, 1NO+1NC aux           | Mitsubishi S-T10 + UT-AX         | 2   |
| 8  | -K3        | Contactors 40 A AC-1, 230 V AC coil, 1NO+1NC aux          | Mitsubishi S-T21 + UT-AX         | 1   |
| 9  | -F1/-F2    | Thermal overload relay 2.5-4 A, class 10A                 | Mitsubishi TH-T18 3.6A           | 2   |
| 10 | -F3        | Thermal overload relay 23-32 A, class 10A                 | Mitsubishi TH-T25 29A            | 1   |
| 11 | -T1        | Control transformer 415/230 V, 500 VA, screened           | IEC 61558-2-6                    | 1   |
| 12 | -G1        | SMPS 230 V AC / 24 V DC, 5 A, 120 W, DIN rail             | Mean Well NDR-120-24             | 1   |
| 13 | -A1        | PLC CPU MELSEC iQ-F FX5U-32MT/ES                          | Mitsubishi FX5U-32MT/ES          | 1   |
| 14 | -A3        | HMI GOT2000 GT2107-WTBD 7" TFT 800x480                    | Mitsubishi GT2107-WTBD           | 1   |
| 15 | -A2        | PID temperature controller 96x96, universal input,        | Yokogawa UT35A / equiv.          | 1   |
| 16 | -B2        | Thermocouple type K, 6 mm x 300 mm, Inconel sheath        | with compensating cable          | 1   |
| 17 | -B3        | Safety high-limit thermostat, 2 x NC, manual reset        | e.g. JUMO STB, 0-400 C           | 1   |
| 18 | -B1        | Position switch, roller lever, 1NO+1NC, IP67              | e.g. Schneider XCKN              | 1   |
| 19 | -KA1...KA4 | Interposing relay 24 V DC coil, 1 c/o 6 A, socket         | e.g. Finder 40.52 + socket       | 4   |
| 20 | -S0        | Emergency stop, 40 mm mushroom, twist release, 2NC        | 22 mm, IP66                      | 1   |
| 21 | -S1...S4   | Push-button 22 mm, flush, 1NO or 1NC                      | green x2 / red x2                | 4   |
| 22 | -S5        | Push-button 22 mm, blue, 1NO                              | lamp test / alarm accept         | 1   |
| 23 | -H1...H6   | Pilot lamp 22 mm LED 24 V DC                              | 2 green, 2 red, 1 amber, 1 white | 6   |
| 24 | -B4        | Alarm hooter 24 V DC, 85 dB                               | 22 mm or surface                 | 1   |
| 25 | -X1        | Terminal block 2.5 mm2 screw / spring, grey               | with end plates and markers      | 45  |
| 26 | -X2        | Terminal block 10 mm2, grey + green/yellow PE             | with end plates and markers      | 12  |
| 27 | -W5        | Ethernet patch cord CAT5e SF/UTP, 1 m, RJ45               | industrial rated                 | 1   |
| 28 | -          | Enclosure 800 x 600 x 250 mm, IP54, RAL 7035              | wall mounting, with gland plate  | 1   |
| 29 | -          | Panel light, 6 A socket, thermostat + filter fan 100 m3/h | -                                | 1   |
| 30 | -A1        | SD memory card 2 GB for the FX5U data logging function    | Mitsubishi NZ1MEM-2GBSD          | 1   |
| 31 | -A3        | GOT battery for the real-time clock and alarm history     | Mitsubishi GT11-50BAT            | 1   |
| 32 | -          | DIN rail, trunking 40x60, earthing bar, labels, ferrules  | consumables                      | 1   |

|                                                   |        |       |            |                        |         |     |
|---------------------------------------------------|--------|-------|------------|------------------------|---------|-----|
| PROJECT<br>OVEN CONTROL PANEL - HEATER + 2 BLOWER |        |       |            | CLIENT / LOCATION<br>- |         |     |
| SHEET TITLE<br>I/O SCHEDULE & BILL OF MATERIALS   |        |       |            |                        |         |     |
| DRAWN                                             | CLAUDE | DATE  | 2026-09-06 | DRAWING No.            | SHEET   | REV |
| CHECKED                                           | -      | SCALE | NTS        | OVN-2026-01            | 13 / 13 | 3   |